

CampusIQ — Complete Tech Stack

A full-stack event management platform for campuses — built with modern web technologies.

Programming & Markup Languages

Technology / Tool	Category	Purpose in CampusIQ
Python	Backend Programming	Powers the entire backend API server — handles business logic, database queries, authentication, payments, PDF generation, and QR code creation.
JavaScript (ES6+)	Frontend Programming	Handles all interactive UI state, event wiring, API communication, QR scanner control, and client-side routing inside the browser.
HTML5	Markup	Provides the base document structure for the single-page React application (index.html entry point).
CSS3	Styling	Powers custom keyframe animations, glassmorphism effects, and gradient tokens via Tailwind utility classes.

Frontend Tech Stack (Vite + React)

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React v19	UI Framework	Core building block for every page and component — Events, Dashboards, Scanner, Profile, EventDetail, etc.
Vite v8	Build Tool & Dev Server	Compiles the JSX/Tailwind source with near-instant HMR locally and creates optimised production bundles.
Tailwind CSS v3	Utility Styling	Utility classes for the blue-indigo-purple gradient theme, responsive grids, dark overlays, and typography.
React Router DOM v7	Client-Side Routing	Manages URL routes and role-based protected paths (/events, /host, /scanner/:id, /edit-event/:id, etc.).
Axios	HTTP Client	Sends authenticated REST requests to the FastAPI backend, attaching Bearer tokens in every header.
Html5-Qrcode	Camera / Hardware API	Opens the device camera and decodes QR codes printed on student ticket images in real time.
Lucide React	Icon Library	Consistent SVG icon set — Camera, Users, Ticket, Trophy, QrCode, ArrowLeft and many more.
React Hot Toast	Notification UI	Non-blocking success/error toast alerts (e.g. 'Checked in: Harshit', 'Invalid Ticket').
React Dropzone	File Upload UI	Handles drag-and-drop for event poster uploads with image preview before form submission.

Backend Tech Stack (FastAPI + Uvicorn)

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FastAPI	Backend Framework	High-performance API framework — exposes all REST endpoints, validates inputs via Pydantic, and manages form-data & file uploads.
Uvicorn	ASGI Server	Asynchronous server that runs the FastAPI application for local development with auto-reload.
Gunicorn	Production Web Server	Production process manager that orchestrates multiple Uvicorn workers on Render (start command: gunicorn -k uvicorn.workers.UvicornWorker main:app).
Pydantic v2	Data Validation	Defines strict schemas for all API request payloads (volunteer applications, payment verification, profile updates).
Python-jose	JWT Authentication	Generates, signs (HS256), and decodes JSON Web Tokens for secure, stateless session management.
Passlib + Bcrypt	Password Security	Hashes user passwords before storing and verifies them safely on login.
ReportLab	PDF Generation	Programmatically builds formatted PDF receipts for successful event fee payments.
Qrcode (python-qrcode)	QR Image Gen	Generates unique ticket QR code PNG images for each registered student, stored on Cloudinary or local disk.
Razorpay SDK	Payment Integration	Creates Razorpay orders and verifies HMAC payment signatures for entry fees and volunteer registration fees.
Cloudinary SDK	Cloud Media Client	Uploads and manages event posters, profile photos, and QR code images on the Cloudinary CDN.

Database & ORM

Technology / Tool	Category	Purpose in CampusIQ
SQLAlchemy v2	ORM	Maps Python model classes (User, Event, Registration, Payment, VolunteerWhitelist) to relational tables and generates optimised SQL.
Alembic	Schema Migration	Tracks schema versions and applies incremental migrations (ALTER TABLE, CREATE TABLE) — critical for zero-downtime production deployments.
SQLite	Local Dev Database	File-based DB (test.db) used during local development. A built-in column-compatibility check in main.py auto-adds missing columns.
PostgreSQL	Production Database	Connected on Render + Supabase via DATABASE_URL env var. Pool settings (pool_size=20, pool_recycle=1800) handle concurrent requests.
psycopg2-binary	PostgreSQL Driver	Binary adapter that allows SQLAlchemy / Python to communicate with PostgreSQL servers.

Cloud Services & External Integrations

Technology / Tool	Category	Purpose in CampusIQ
Cloudinary	Cloud File Storage	Stores all media assets (event posters, user profiles, QR code tickets) statlessly, returning CDN secure_url for serving.
Razorpay	Payment Gateway	Handles INR payment flows: order creation → client checkout → server-side signature verification → registration confirmation.
Supabase	Managed PostgreSQL DB	Free-tier PostgreSQL hosting used as the production database; connected via a DATABASE_URL connection string.
Render	Backend Hosting Platform	Deploys the FastAPI backend as a web service. Runs alembic upgrade head on startup and serves via Gunicorn.
Vercel / Netlify	Frontend Hosting	Builds and serves the React+Vite static bundle. VITE_API_URL env var points the frontend to the Render backend.